LEARNING DYNAMICS

If neural networks can genuinely be understood mechanistically, in terms of circuits, it seems like there almost *has to* be some way to understand the learning process in terms of the dynamics of circuits changing. Induction heads offer an interesting preliminary bridge between these topics, and are a source of optimism for such a connection. This section will briefly review some strands of work on the learning dynamics side, which seem particularly promising to think about if one wanted to pursue such a connection further.

One remarkable result in learning dynamics, by Saxe *et al* [23], has been the discovery of *closed form* solutions to learning dynamics for linear neural networks without activation functions. The exciting thing about this work is that it actually provides a simple way to conceptually think about neural network learning in a simplified case. (In follow up work, Saxe *et al* also explore connections between this framework and models learning to represent semantic information [36].) We're unable to provide a detailed review of this work here, but we note that Saxe *et al*'s framework could naturally suggest a circuit lens for thinking about learning dynamics. Very roughly, they find that linear neural networks can be understood in terms of the evolution of independent paths through the network, with each path corresponding to a principal component of the data. These paths might be thought of as circuits.

Another interesting line of work has been the study of the geometry of neural network loss surfaces (*e.g.* [37, 38, 39, 40]). Here, our thoughts on the connection are more superficial, but it seems like there must be some way in which aspects of the loss surface connect to the formation of circuits. Very concretely, it seems like the phase change we've described in this paper must correspond to some very large feature in the loss landscape of transformers.

UNIVERSALITY

In the context of interpretability and circuits, "universality" [41] or "convergent learning" [42] is when multiple models develop the same features and circuits. Universality might seem like an intellectual curiosity, but the circuits thread argues that universality plays a critical role in what kind of interpretability makes sense:

> *[I]magine the study of anatomy in a world where every species of animal had a completely unrelated anatomy: would we seriously study anything other than humans and a couple domestic animals? In the same way, the universality hypothesis determines what form of circuits research makes sense. If it was true in the strongest sense, one could imagine a kind of "periodic table of visual features" which we observe and catalogue across models. On the other hand, if it was mostly false, we would need to focus on a handful of models of particular societal importance and hope they stop changing every year.* [41]

Research on universality began with *Li et al.* [42] who showed that many neurons are highly correlated with neurons in retrained versions of the same model. More recently, a number of papers have shown that in aggregate, neural networks develop representations with a lot of shared information (*e.g.* [43, 44]). The Circuits thread tried to extend this notion of universality from features to *circuits*, finding that not only do at least some families of well-characterized neurons reoccur across multiple networks of different architectures and that the same circuits [41], but the same circuits appear to implement them [45].

Certain kinds of universality are often implicitly assumed in the language model attention head interpretability literature. For example, it seems widely accepted that "previous token" attention heads form across many transformer language models (*e.g.* [26, 27]). The implicit hypothesis of universal attention heads – that is, attention heads with the same attention patterns in different models – isn't exactly the same thing as the kind of feature universality studied in the vision context, but is kind of analogous.

Our work in this paper has analogies to many of these strands of prior work. Like the previous attention head papers, we describe the induction head pattern as a universal attention pattern. However, our analysis of these heads' OV and QK circuits extends this claim of universality to the circuit level, similar to the original Circuits thread. And a corollary of our analysis of the OV circuit is a claim about what feature the attention head computes (roughly: the token embedding of the token following a previous copy of the present token) which is more similar to the traditional work on universality.

Separate from all of this, it's worth mentioning that increasingly there's evidence for a particularly extreme kind of universality at the intersection of neuroscience and deep learning. Increasingly, research suggests that biological and artificial neural networks learn similar representations (*e.g.* [46, 47, 48]). In fact, Goh *et al.* [49] find that multimodal "concept" neurons found in humans (such as the famous "Jennifer Anniston neuron") occur in neural networks.

ATTENTION PATTERNS IN TRANSLATION-LIKE TASKS

In Argument 4, we saw an induction head that helps implement translation. Although we're not aware of anything quite so general in the prior literature, there are reports of attention patterns which, in retrospect, seem somewhat similar. Often, in translation-like tasks, we see attention attend to the token which is *about to be translated*. We see this in literal translation (*e.g.* [50]) and also in voice recognition (*e.g.* [51] where the model attends to the portion of the audio about to be transcribed). Visualizations of this in the encoder-decoder context often slightly obscure the induction-like nature of the attention patterns, because the decoder is visualized in terms of the output tokens predicted per time step rather than its input tokens.

Comments & Replications

Inspired by the original Circuits Thread and Distill's Discussion Article experiment, the authors invited several external researchers who were also investigating induction heads to comment on this work. Their comments are included below.

REPLICATION

Adam Scherlis *is a researcher at Redwood Research.*

Redwood Research has been working on language model interpretability, inspired in part by Anthropic's work. We've found induction heads reliably appearing in two-layer attention-only transformers. Their structure roughly matches the description in "Analyzing a Two-Layer Model" from Anthropic's previous paper, with previous-token heads in layer 0 and induction heads in layer 1 (often one of each). These each have the expected attention behaviors. We tested this by replacing attention-score matrices with idealized versions and comparing the change in loss to the change from ablating that head. Replacing the previous-token head's scores with exact previous-token attention recovered 99% of the loss difference. Replacing the induction head's scores with a simple approximation (attending to the first token and to exact `[A][B]...[A]` matches) recovered about 65% of the loss difference. Our induction heads also match patterns of the form `[A][B][C]...[A][B]→[C]`; including this in the substituted attention scores recovered an additional 10% of loss difference. The induction head's OV circuit copies tokens, including some fuzzy matching of semantically similar tokens. Its QK circuit is dominated by K-composition with the previous-token head; the previous-token head's OV matrix copies information to a new subspace and the induction head's QK matrix copies it back to the usual token embedding. We've also cataloged a few other kinds of attention heads, including skip-trigram heads.

REPLICATION

Tom Lieberum *is a master's student at the University of Amsterdam.*

I used the empirical criterion for induction heads presented in this paper to look for induction heads in publicly available models. To reiterate: on a sequence of tokens `[A][B] .... [A] → [B]`, a head is called an induction head if it attends to the previous `[B]` when reading the last `[A]`, and if the previous `[B]` increased the logit of the last `[B]`.

Under this definition, I found potential induction heads in GPT2 and GPT-Neo mostly starting in the mid-depth region. I made an interactive version to explore attention and logit attribution across all layers and heads on the long Harry Potter prompt with the repetition of the first paragraph. It can be accessed here. For example, for GPT2-XL, head 20 in layer 21 seems to be an induction head, as well as head 0 in layer 12 of GPT-Neo-2.7B. For these heads we can see that virtually every token in the repetition of the first paragraph attends to its following token in the original paragraph. Thanks to EleutherAI for providing the compute resources for this project.

At one point the paper speculates on the minimal context length necessary to form induction heads. On a synthetic dataset, I have already found induction to be easily learned with a context length of 4. However, I believe that this is, in general, a property of the dataset/data distribution rather than the model itself, i.e. "how useful is induction for this task, on this dataset?". While it is intriguing to think about the statistics of language in this way, I am not sure how useful this line of research would be to the overall project of interpretability.